Kei Imada

Pronounced like the letter after "J"

Email: kei (at) cs (dot) cornell (dot) edu • **Github**: keikun555 • **Website**: keikun555.github.io • **LinkedIn**: kei-imada

ABSTRACT

Fourth year Ph.D. candidate at **Cornell University**, developing discrete verification tools that prove temporal, probabilistic properties of continuous-time systems. Former **networking software engineer**.

Skills: Python, Golang, C, C++, OCaml, LaTeX, Z3, TLA+, MPI, CUDA, React, Typescript, SQL, Japanese, and Chinese.

PUBLICATIONS

A Hoare Logic for Continuous-Time Stochastic Processes *In Submission*

Kei Imada and Justin Hsu

A Nonstandard Temporal Logic for Continuous-Time Verification *In Submission*

Kei Imada, Kazuki Watanabe, Justin Hsu, and Kohei Suenaga

ACADEMICS

Cornell University in *Ithaca, NY* **August 2023 → Present**

Doctor of Philosophy in Computer Science *Phi Kappa Phi*, *4.1 GPA*, **Advisor**: Justin Hsu

• Kleene Algebra, Category Theory, Mathematical Logic, Compilers, Program Synthesis, Formal Methods in Robotics

Swarthmore College in *Swarthmore, PA* **August 2016 → May 2020**

Bachelor of Arts with dual majors in Computer Science and Mathematics *Cumulative 3.94 and Major 3.95 GPA*

Thesis: The Equivalence of Typed λ Calculi and Cartesian Closed Categories *Phi Beta Kappa and Sigma Xi*

• Parallel and Distributed Computing, Programming Languages, Algorithms, Networks, Artificial Intelligence, Natural Language Processing, Category Theory, Topology, Real Analysis, Modern Algebra, Differential Equations, Modeling

Budapest Semesters in Mathematics in *Budapest, Hungary* **January 2019 → May 2019**

Magas Kitüntetéssel High Honors (A or above on 5+ courses) *4.0 GPA*

• Real Functions and Measures, Theory of Computing, Conjecture & Proof, Topology, Mathematical Cryptography

EXPERIENCE

Teaching Assistant at *Cornell University* in Ithaca, NY **August 2025 → May 2026**

CS 3410: Computer System Organization and Programming **Spring 2026**

- Led weekly lab sessions to clarify class material on RISC-V, memory management, and concurrency
- Monitored and coached students through Ed Discussions, an online discussion platform

CS 6120: Advanced Compilers **Fall 2025**

- Instructed 30 students on common compiler optimizations, garbage collection, and data flow analysis
- Monitored and coached students through Zulip, an online discussion platform
- Graded biweekly assignments and provided feedback on code and design choices

Research Assistant at *Cornell University* in Ithaca, NY **August 2023 → Present**

Research Intern at *National Institute of Informatics* in Tokyo, Japan **February 2025 → August 2025**

• Collaborated with Ichiro Hasuo and Kazuki Watanabe on developing a program logic for continuous-time verification

• Presented preliminary results at CSCAT 2025 and Kyoto University

Software Engineer at *Pure Storage* in Mountain View, CA **August 2020 → August 2023**

• Developed an internal firewall library with in-memory caching and batch-commit functionalities for performance

• Improved an existing Python *iptables* library's performance by 2 orders of magnitude

• Integrated *iptables* into FlashBlade clusters and improved its initialization and sync performance

• Implemented VLAN translation with field processor rules into Broadcom Tomahawk3 chips

• Spearheaded a formal verification project to secure FlashBlade's networking stacks

• Triaged over 500 test failures and discovered zero-day vulnerabilities and race conditions

• Organized and presented a 60-minute tech talk on testbed architectures and connectivity failures for over 80 engineers

Software Engineer Intern at *Pure Storage* in Mountain View, CA **June 2019 → August 2019**

• Designed and implemented a scalable web analytics tool that detects and diagnoses SSD drive failures

• Improved latency by 2-fold with Redis caching layers storing structured responses from Amazon Redshift

• Developed the frontend using React and Typescript, with *ag-grid*, *highcharts*, and *react-select* as core components

Project Lead at *Swarthmore College Computer Society* (SCCS) **January 2018 → May 2020**

• Developed a website that streamlined carpooling between Swarthmore and popular transportation hubs

• Scheduled more than 200 rides which saved over $2,000 in transportation costs

• Awarded a $5,000 scholarship by developing a website to assist students in scheduling their courses

SINGING

Repertoire: My Way, Fly Me To The Moon, My Funny Valentine, Unchained Melody, 千里之外 , and more

• Awarded a semesterly $560 scholarship to take classical singing lessons with Professor Nancy Jantsch

• A high baritone at Swarthmore College Choir for 7 semesters